SINK OR FLOAT?

A Complete Discovery Guide & Activity Book on Density

Science Learning for Grade 2

Core Science Concepts

1. What is Sinking?

When something is heavy for its size, it goes down to the bottom of the water. We call this **sinking**! The scientific word for being heavy for your size is **dense**. Dense objects are packed with lots of matter in a small space, so they are heavier than the same amount of water and down they go!

Examples: Rocks, metal coins, keys, glass marbles, metal spoons, pebbles, bolts, and nails.

2. What is Floating?

When something floats, it stays on top of the water. Objects that are light for their size are **less dense** than water. Because they weigh less than the same amount of water, water pushes them up and they rest on the surface just like a leaf resting gently on a pond!

Examples: Leaves, corks, wooden blocks, plastic toys, rubber ducks, empty bottles, feathers, and sponges.

3. The Big Word: DENSITY!

Density means how much matter (stuff) is packed into a certain space. Imagine two boxes that are exactly the same size. One is filled with feathers and one is filled with rocks. The rock box is much heavier — it is *more dense*. The feather box is lighter — it is *less dense*.

Water has its own density. If something is more dense than water → it sinks. If something is less dense than water → it floats. That's the whole secret!

Density Quick Guide

More Dense Than Water (SINKS)	Less Dense Than Water (FLOATS)
Rock, Coin, Key, Marble	Wood, Cork, Leaf
Metal Spoon	Plastic Toy, Feather

Key Science Vocabulary

Term	Definition
Sink	To go down to the bottom of water because the object is more dense than water.
Float	To stay on top of water because the object is less dense than water.
Density	How much matter (stuff) is packed into a certain space. Dense = heavy for its size.
Dense	Having a lot of matter packed tightly together; heavy for its size.
Less Dense	Having less matter packed together; light for its size; these objects float.
Predict	To make a smart guess about what will happen before you test it.
Observe	To look carefully and notice what is happening during an experiment.
Experiment	A test we do to find out whether our prediction was right or wrong.

Student Observation Worksheet

Name: ________________________ **Date:** ________________

☑ Instructions

1. Pick an object.
2. Hold it in your hand — does it feel heavy or light?
3. **PREDICT:** Will it sink or float? Write your prediction.
4. Place the object gently in the water and observe.
5. Record the **RESULT** and check if your prediction was correct!

Object	Prediction (Sink / Float)	Result (Sink / Float)	Was I right? (Yes / No)
1.			
2.			
3.			
4.			
5.			
6.			
7.			
8.			

Reflection Questions

1. Which object surprised you the most? Why?

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2. What do you think makes something sink instead of float?

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3. Can you think of a real-life example where knowing if something sinks or floats is important?

Comprehension Quiz

Q1. What happens to an object that is MORE dense than water?

- It floats to the top.
- It sinks to the bottom.
- It disappears.
- It turns into ice.

Answer: It sinks to the bottom.

Q2. Which of the following objects will most likely FLOAT?

- A metal coin
- A rock
- A wooden block
- A glass marble

Answer: A wooden block

Q3. What does the word 'density' mean?

- The colour of an object
- How much stuff is packed into a space
- How fast an object moves
- The temperature of water

Answer: How much stuff is packed into a space

Q4. A leaf rests on top of a pond. This means the leaf is...

- More dense than water
- The same density as water
- Less dense than water
- Made of water

Answer: Less dense than water

Q5. Before testing an object in water, a good scientist should...

- ○ Just throw it in
- ○ Make a prediction first
- ○ Ask a friend to do it
- ○ Skip the experiment

Answer: Make a prediction first

Extension Activities

🏠 At-Home Challenge

Go home and test 5 objects from your kitchen or bathroom! Draw each object and record whether it sinks or floats. Bring your findings to share with the class tomorrow.

✍️ Creative Writing Prompt

Imagine you are a tiny submarine travelling underwater. Write 3-4 sentences describing what you see sinking to the bottom and what you see floating above you!

🎨 Art Activity

Draw a large glass of water. On the left side, draw 3 objects that sink and show them at the bottom. On the right side, draw 3 objects that float and show them at the top. Label each object and write 'dense' or 'less dense' next to it.

🚢 Real-World Connection

Big steel ships float even though steel is heavy! This is because the ship's shape traps air inside, making the whole ship less dense than water overall. *Try it yourself:* If we crumple a piece of aluminium foil into a ball, does it sink? What if we shape it like a boat?